

Question 1

If $\psi_{n,\mathbf{k}}(\mathbf{r})$ is a Bloch wavefunction,

(a) Show how $\psi_{n,\mathbf{k}}(\mathbf{r})$ relates to $\psi_{n,\mathbf{k}+\mathbf{G}}(\mathbf{r}+\mathbf{R})$, where $\mathbf{R}$ and $\mathbf{G}$ are arbitrary translation vectors of the real and reciprocal lattices, respectively. (2)

(b) Explain how $\epsilon_n(\mathbf{k})$ relates to $\epsilon_n(\mathbf{k}+\mathbf{G})$. (1)

Solution**(a) Bloch theorem**

According to Bloch's theorem,

$$\psi_{n,\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n,\mathbf{k}}(\mathbf{r}), \quad (1)$$

where

$$u_{n,\mathbf{k}}(\mathbf{r}+\mathbf{R}) = u_{n,\mathbf{k}}(\mathbf{r}).$$

Therefore,

$$\psi_{n,\mathbf{k}}(\mathbf{r}+\mathbf{R}) = e^{i\mathbf{k}\cdot(\mathbf{r}+\mathbf{R})} u_{n,\mathbf{k}}(\mathbf{r}+\mathbf{R}) \quad (2)$$

$$= e^{i\mathbf{k}\cdot\mathbf{R}} \psi_{n,\mathbf{k}}(\mathbf{r}). \quad (3)$$

Hence,

$$\boxed{\psi_{n,\mathbf{k}}(\mathbf{r}+\mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{R}} \psi_{n,\mathbf{k}}(\mathbf{r})} \quad (4)$$

For a reciprocal-lattice vector $\mathbf{G}$,

$$\mathbf{G} \cdot \mathbf{R} = 2\pi m, \quad m \in \mathbb{Z},$$

so

$$e^{i\mathbf{G}\cdot\mathbf{R}} = 1.$$

Thus $\mathbf{k}$ and $\mathbf{k}+\mathbf{G}$ represent equivalent points in reciprocal space. Therefore,

$$\boxed{\psi_{n,\mathbf{k}+\mathbf{G}}(\mathbf{r}+\mathbf{R}) = e^{i\mathbf{k}\cdot\mathbf{R}} \psi_{n,\mathbf{k}}(\mathbf{r})} \quad (5)$$

up to an overall phase convention.

(b) Energy periodicity

The energy bands are periodic in reciprocal space:

$$\boxed{\epsilon_n(\mathbf{k}+\mathbf{G}) = \epsilon_n(\mathbf{k})} \quad (6)$$

Question 2

For a 1-dimensional semiconductor, the ratio of the hole and electron masses is

$$\frac{m_h}{m_e} = 1.25.$$

How far above the mid-gap (middle of the bandgap) does the intrinsic Fermi level lie at room temperature?

Given,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

(6)

Solution

For an intrinsic semiconductor,

$$n = p.$$

The electron and hole concentrations are

$$n = N_c e^{-(E_c - E_i)/(k_B T)}$$

and

$$p = N_v e^{-(E_i - E_v)/(k_B T)}.$$

Therefore,

$$E_i = \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \left(\frac{N_v}{N_c} \right).$$

Let

$$E_{\text{mid}} = \frac{E_c + E_v}{2}.$$

Then,

$$E_i - E_{\text{mid}} = \frac{k_B T}{2} \ln \left(\frac{N_v}{N_c} \right).$$

For a 1D semiconductor,

$$N_c \propto \sqrt{m_e}, \quad N_v \propto \sqrt{m_h}.$$

Hence,

$$\frac{N_v}{N_c} = \sqrt{\frac{m_h}{m_e}}.$$

Therefore,

$$E_i - E_{\text{mid}} = \frac{k_B T}{2} \ln \left[\sqrt{\frac{m_h}{m_e}} \right] \quad (7)$$

$$= \frac{k_B T}{4} \ln \left(\frac{m_h}{m_e} \right). \quad (8)$$

At room temperature,

$$T = 300 \text{ K}$$

and

$$k_B = 8.617 \times 10^{-5} \text{ eV/K}.$$

Thus,

$$k_B T = (8.617 \times 10^{-5})(300) = 0.025851 \text{ eV}.$$

Therefore,

$$E_i - E_{\text{mid}} = \frac{0.025851}{4} \ln(1.25) \quad (9)$$

$$= 0.001441 \text{ eV} \quad (10)$$

$$= 1.44 \text{ meV}. \quad (11)$$

Question 3

Argue that an element with 3 valence electrons per atom, crystallizing in a simple cubic crystal structure, cannot be a semiconductor.

Could it be a semiconductor if it crystallized in the FCC structure?

(1.5 + 1.5)

Solution**(a) Simple Cubic Structure**

A simple cubic (SC) lattice has one atom per primitive unit cell. If each atom contributes 3 valence electrons, there are

3 electrons per primitive cell.

Each energy band can accommodate a maximum of 2 electrons per primitive cell due to spin degeneracy.

For a semiconductor or insulator, the occupied bands must be completely filled, requiring an even number of electrons per primitive cell:

$$2, 4, 6, \dots$$

With 3 electrons, the first band contains 2 electrons and one electron remains to occupy the next band. Hence, the next band is partially filled.

A partially filled band places the Fermi level inside a band rather than in a band gap.

Therefore, it cannot be a conventional semiconductor. (13)

(b) FCC Structure

An FCC conventional unit cell contains 4 atoms, giving

$$4 \times 3 = 12$$

valence electrons per conventional unit cell.

However, band filling must be considered using the primitive unit cell. An FCC lattice contains one atom per primitive unit cell. Therefore, there are again

3 valence electrons per primitive cell.

The odd number of electrons per primitive cell remains unchanged. Thus the same band-filling argument applies.

It cannot become a conventional semiconductor merely by crystallizing in the FCC structure. (14)

Question 4

The intrinsic carrier concentration of an FCC semiconductor at room temperature, with an indirect band gap

$$E_g = 0.66 \text{ eV}$$

and relative dielectric constant κ is

$$n_i = 3 \times 10^{13} \text{ cm}^{-3}.$$

Effective mass measurement is performed in a magnetic field B .

The remaining portion of Question 4 is not visible in the supplied photographs.

Solution

The supplied photographs do not contain the complete statement of Question 4. The numerical value of κ , the magnetic-field value, and the remainder of the question are cut off.

Therefore, the complete solution cannot be obtained reliably without guessing the missing information.

Upload the remaining portion of Q4 to complete its solution.

Quick Revision – Final Answers

Q.	Final Result
1(a)	$\psi_{n,\mathbf{k}}(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k} \cdot \mathbf{R}} \psi_{n,\mathbf{k}}(\mathbf{r})$. $\mathbf{k}$ and $\mathbf{k} + \mathbf{G}$ are equivalent.
1(b)	$\epsilon_n(\mathbf{k} + \mathbf{G}) = \epsilon_n(\mathbf{k})$
2	$E_i - E_{\text{mid}} \approx 1.44 \text{ meV}$ above mid-gap.
3	3 valence electrons per primitive cell give odd band filling; neither SC nor FCC alone removes this problem.
4	Incomplete in supplied photographs.

End of Solutions